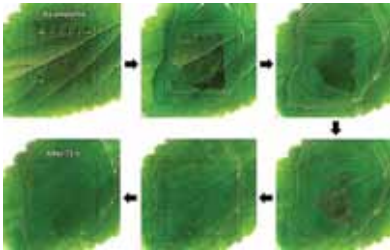




High-performance biodegradable memory device

Researchers at KIST (Seoul, South Korea) have developed a high-performance, fully biodegradable memory device utilising PCL-TEMPO, a hybrid of polycaprolactone and a



A step-by-step demonstration showing how the device, placed on a green leaf, completely dissolves in water within 72 hours (Credit: Korea Institute of Science and Technology)

data-storing organic compound. It offers durable data retention across over a million cycles and survives 3000 bends, yet dissolves completely in water within three days. Ideal for disposable wearables and implants, it

eliminates e-waste and secondary surgeries. With tunable degradation, it is safe for the body and perfect for medical, military, and eco-friendly tech. Future upgrades may include light-responsive, self-healing features.

Compact RF switch powers 6G and autonomous tech

Researchers at UNIST (Ulsan, South Korea) have developed a memristor-based RF switch using vanadium oxide, specifically designed for 6G and autonomous technologies. Its nonvolatile design retains states with zero standby power, which is ideal for always-on devices. With nanosecond switching and operation up to 67GHz (and simulated 4.5THz), it offers low loss and high isolation. Integrated into a tunable filter, this compact switch is set to power agile, high-speed, energy-efficient RF front-ends of the future.

Ultrasonic test system detects hidden defects in batteries

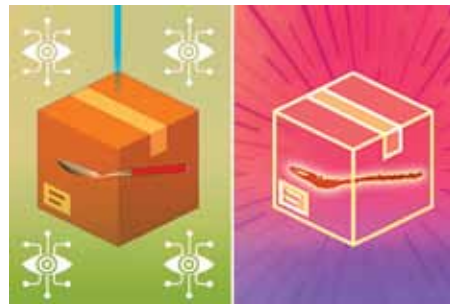
Drexel University (Philadelphia, Pennsylvania) has developed a compact ultrasonic testing system that inspects live batteries in real time, without causing damage, delays, or the use of X-rays. Using scanning acoustic microscopy, it reveals hidden flaws, such as cracks and gas pockets, by tracking the behaviour of sound waves. The low-cost setup fits on a bench and includes open source software for automation and fast analysis. It is quicker, safer, and more scalable, making it ideal for electric vehicle labs and factories. This breakthrough enhances quality control and accelerates next-generation battery development by identifying issues before they arise.

Power and data delivery made possible by laser light through fibre cables

Researchers at the University of Ottawa (Canada) have redefined long-range connectivity using photonic power conversion. By sending laser light through standard fibre optic cables, they can now deliver both power and data efficiently over distances exceeding 1km, even in extreme environments. Due to the development of multi-junction semiconductor converters at SUNLAB, an efficiency of over 53% was achieved at telecom wavelengths. This enables practical deployment in smart grids, aircraft, IoT, and underwater systems, without requiring new power infrastructure.

Radar-based system brings 'X-Ray vision' to robots

MIT (Cambridge, Massachusetts) researchers have unveiled mmNorm, a radar-based system that uses mmWave signals to reconstruct hidden 3D objects with 96% accuracy,



A new system enables a robot to use reflected Wi-Fi signals to identify the shape of a 3D object that is hidden from view, which could be especially useful in warehouse and factory settings (Credit: MIT News; Courtesy of the researchers)

significantly outperforming existing technology. Mounted on a robotic arm, it estimates surface normals to generate detailed 3D models of concealed items, even through packaging.

From factories to homes and airports, mmNorm enables smart robots to detect and interact with occluded objects, transforming automation, security, and AR. Supported by the NSF, Microsoft, and the MIT Media Lab, this is next-generation robotic vision.

Novel 3D printing method draws ultra-thin, flexible, conductive circuits

A team at Dalian University of Technology (Dalian, China) has developed tension-driven fluid drawing, a novel 3D printing method that pulls silver nanoparticle ink into fine threads, rather than pushing it. With just one heated needle, it draws ultra-thin, flexible, and conductive circuits without the need for layer stacking or vias. This precision method delivers strong durability across surfaces, surviving 200+ bends. It is ideal for wearable medical sensors, compact IoT devices, and stretchable displays.